

TidyTuesday Week 28

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Wednesday 8th July, 2026

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1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

This week the random series generator served up a genuinely bizarre pairing: the Producer Price Index for Frozen Specialty Food Manufacturing (think frozen dinners and specialty entrees, priced against a December 1982 baseline) and the total carbon dioxide emissions from petroleum used in U.S. transportation. On paper these two have absolutely nothing to do with each other – one tracks the wholesale cost of your freezer aisle, the other tracks tailpipe emissions from the nation's cars, trucks, and planes. Naturally, that made it the perfect candidate for our little correlation lottery.

And wouldn't you know it, the numbers actually lined up surprisingly well. The regression came back with an R-squared of about 0.58, meaning the frozen food price index "explains" nearly 60 percent of the variation in transportation petroleum emissions, with a p-value small enough (around one in ten million) to make a real economist raise an eyebrow. The slope suggests that every one-point rise in the frozen food index goes hand in hand with roughly seven more million metric tons of CO₂. If you squinted, you could almost write a headline: "Frozen Pizza Prices Predict America's Carbon Footprint."

Of course, you should absolutely not do that. The far more boring explanation is that both series simply drifted upward over the same decades – prices tend to climb with inflation, and transportation emissions climbed for years as the economy grew. Two things that both go up over time will happily produce an impressive-looking correlation even when there's no connection whatsoever. The software also flagged a large condition number, which is its polite way of hinting that the tidy fit shouldn't be taken too seriously.

As always, this is the whole point of the project: we pair series at random and see what falls out, and spurious correlation is the rule here, not the exception. There's no causal story, no rigorous claim, and certainly no reason to reconsider your grocery budget on climate grounds. It's just a fun reminder that with enough time series lying around, you can make almost anything look related. See you next week for whatever odd couple the generator dredges up next.

2 Date: 2026-07-08

Series ID: PCU311412311412P

This series is titled Producer Price Index by Industry: Frozen Specialty Food Manufacturing: Primary Products and has a frequency of Monthly. The units are Index Dec 1982=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1982-12-01 and the observation end date is 2026-05-01. The popularity of this series is 1.

Series ID: EMISSCO2TOTVTCPEA

This series is titled Transportation Carbon Dioxide Emissions, Petroleum for United States (DISCONTINUED) and has a frequency of Annual. The units are Million Metric Tons CO2 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1980-01-01 and the observation end date is 2017-01-01. The popularity of this series is 3.

2.1 Regression Tables and Plots

Dep. Variable:	value_fred_EMISSCO2TOTVTCPEA	R-squared:	0.578			
Model:	OLS	Adj. R-squared:	0.565			
Method:	Least Squares	F-statistic:	45.15			
Date:	Wed, 08 Jul 2026	Prob (F-statistic):	1.18e-07			
Time:	14:38:29	Log-Likelihood:	-215.74			
No. Observations:	35	AIC:	435.5			
Df Residuals:	33	BIC:	438.6			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	698.8159	150.774	4.635	0.000	392.063	1005.568
value_fred_PCU311412311412P	7.0696	1.052	6.720	0.000	4.929	9.210
Omnibus:	4.673	Durbin-Watson:	0.183			
Prob(Omnibus):	0.097	Jarque-Bera (JB):	4.246			
Skew:	0.788	Prob(JB):	0.120			
Kurtosis:	2.346	Cond. No.	1.08e+03			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 1.08e+03. This might indicate that there are strong multicollinearity or other numerical problems.

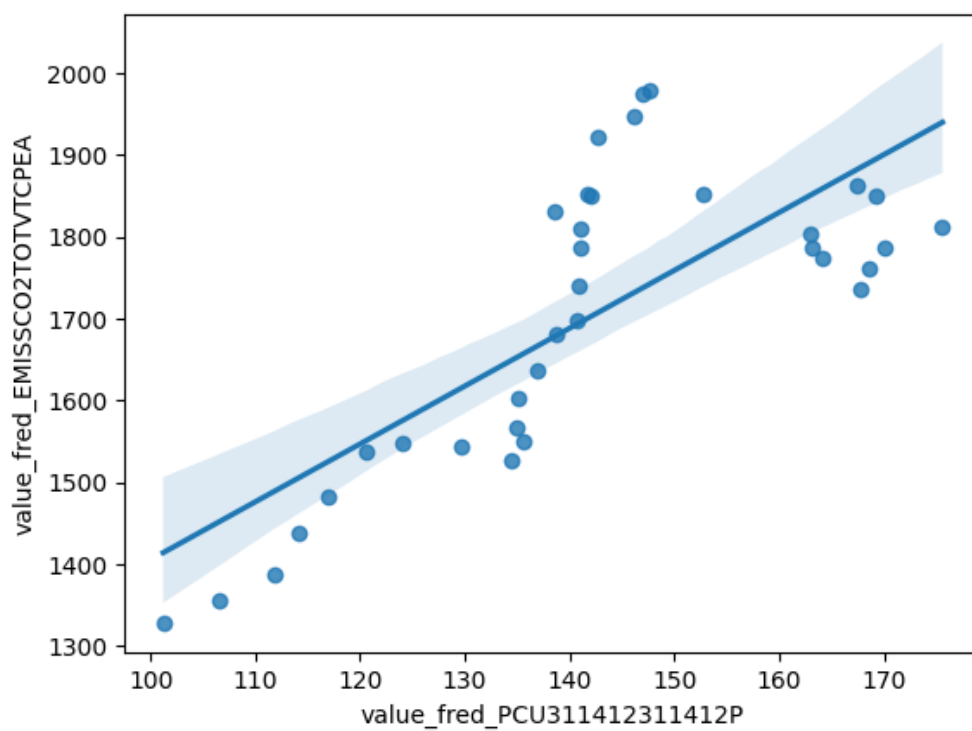


Figure 1: Regression Plot for 2026-07-08